PROGRAM

```
# Facts
facts = {
   "attendance_low"
}

# Rules
rules = [
   ("attendance_low", "student_at_risk"),
   ("student_at_risk", "counseling_required"),
   ("counseling_required", "advisor_notification")
]

def forward_chaining(facts, rules):
   inferred = set(facts)
   changed = True
   while changed:
      changed = False
      for condition, conclusion in rules:
         if condition in inferred and conclusion not in inferred:
            inferred.add(conclusion)
            changed = True
   return inferred
result = forward_chaining(facts, rules)
print("Initial Facts:")
for fact in facts:
   print("-", fact)

print("\nInferred Facts:")
for fact in result:
   print("-", fact)

print("\nFinal Recommendation:")

if "advisor_notification" in result:
   print("Student is at risk. Counseling is required.")
   print("Advisor notification is generated.")
```

OUTPUT

Initial Facts:
- attendance_low

Inferred Facts:
- attendance_low
- student_at_risk
- counseling_required
- advisor_notification

Final Recommendation:
Student is at risk. Counseling is required.
Advisor notification is generated.